

CST8506 - Lab 2

Support Vector Machines

Due Date: Check Brightspace for due dates.

Introduction

The goal of this lab is to classify Wine dataset using SVM (using standardized dataset, transformed dataset by applying PCA and LDA). The dataset can be found in sklearn's datasets package (https://scikit-learn.org/stable/modules/generated/sklearn.datasets.load_wine.html).

Steps (all these steps should be done in Python):

1. Load the dataset.
2. Print number of instances, number and names of attributes and first 5 rows of the dataset.
3. Split the dataset into train and test sets.
4. Standardize data in the dataset.
5. Fit SVM model on the train set and make predictions for the test set. (use linear, polynomial and rbf kernels)
6. Apply PCA to find the principal components of the standardized dataset. Find the best 'd'.
7. Using the first 2 components, plot 2D graphs –the train set (color-coded by each class) and the support vectors too (edge-colors should be color-coded by each class) for all three models (refer to the slides for a reference) for **each** SVM model.
8. Apply LDA to reduce the dimensionality of the standardized dataset. Mention how many components you have used.
9. For the new dataset, plot 2D graphs –the train set (color-coded by each class) and the support vectors too (edge colors should be color-coded by each class) for all three models (refer to the slides for a sample) for each SVM model.
10. Find accuracies and CM for all three models on Standardized dataset, reduced dataset by PCA, and reduced dataset by LDA **and tabulate them in Python**. The table should be in the following format (For PCA and LDA, **fill in the value for d**, which is the number of components):

	Accuracy	Confusion Matrix
Standardized, linear SVM		
Standardized, poly SVM		
Standardized, RBF SVM		
PCA (d), linear SVM		
PCA(d), poly SVM		
PCA(d), RBF SVM		
LDA(d), linear SVM		
LDA(d), poly SVM		
LDA(d), RBF SVM		

To get grades,

1. You should be ready with your Python code and results.
2. Submit your Colab notebook file and the answer document to Brightspace.
3. In each graph, you will have 6 colors – 3 classes and 3 sets of support vectors.